

with each other through the GUI, such as:

GUI. It is used to create schematics, set up simulations, display simulation results and to write VHDL code.

Back-end analogue simulator. The analogue simulator is a command line program which is run by the GUI in order to simulate the schematic that you previously setup. It takes a netlist, checks it for errors, performs the required simulation actions and finally produces a dataset.

Simple text editor. This is used to display netlists and simulation logging information, and also to edit files included by certain components (like Spice netlists or Touchstone files).

Filter synthesis application. The program can be used to design various types of filters.

Transmission line calculator. This can be used to design and analyse different types of transmission lines (examples: micro-strips, coaxial cables).

Component library. The component library manager holds models for real-life devices, such as transistors, diodes, bridges and op-amps. The user can extend it with additional components.

Attenuator synthesis application. The program can be used to design various types of passive attenuators.

Command line conversion program. The conversion tool is used by the GUI to import and export datasets, netlists and schematics from and to other CAD/EDA software. The supported file formats as well as usage information can be found on the home page of Qucs.

Additionally, the GUI steers other EDA tools. For digital simulations (via VHDL) the program FreeHDL (see <http://www.freehdl.seul.org>) is used. And for circuit optimisations ASCO (see <http://asco.sourceforge.net>) is configured and run.

Getting started

Once the software has been successfully installed on your system you can start it by issuing the `# qucs` command, or by clicking the appropriate icon on

your start menu or desktop. Qucs is a multi-lingual program. Therefore, depending on your system's language settings, the Qucs GUI appears in dif-

ferent languages.

On the left hand side you will find the Projects folder opened, as shown in Fig. 1. Usually, the projects folder will be empty if you are using Qucs for the first time. The large area on the right hand side is the schematic area. Right above it you can find the menu bar and the toolbars. Go to File → Application Settings menu to configure the language and appearance of Qucs.

For the language and font settings to take effect, the application (as seen in Fig. 2) must be closed either via the Ctrl + Q shortcut or the File → Exit menu entry. Then re-start Qucs.

Setting up schematics

The following sections will enable the user to set up some simple schematics. For this, first create a new project named 'WorkBook.' Either press the New button above the projects folder or use the menu entry Project → New Project, and enter the new project name as shown in Fig. 3.

Confirm the dialogue by pressing the Create button. When done, the project is opened and Qucs switches to the Content tab.

In the Content tab, as seen in Fig. 4, you will find all the data related to the project. It contains your schematics, the VHDL files, data display pages, datasets as well as any other data (like datasheets). On the right hand side, an untitled and empty schematic window is displayed. Now you can start to edit the schematic. The available components can be found in the Components tab.

The window in Fig. 5 is displayed on clicking the Components tab. There are lumped components (such as resistors and capacitors), sources (such as DC and AC

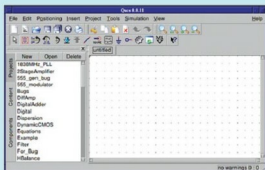


Fig. 4: Qucs has been started

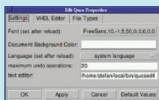


Fig. 5: Application Setting Dialog Box

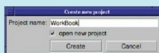


Fig. 6: New Project Dialog

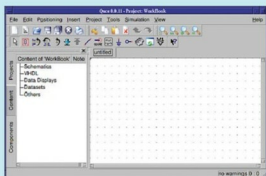


Fig. 7: New empty project has been created

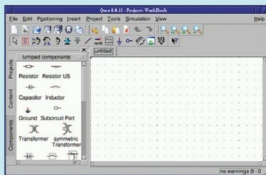


Fig. 8: Components tab